AME-Wk3-ODEs-2026_Annotation_updated

Wednesday, 25 February 2026 1:55 PM

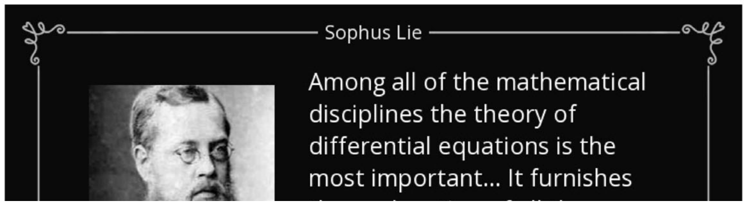
AME-Wk3-ODEs-

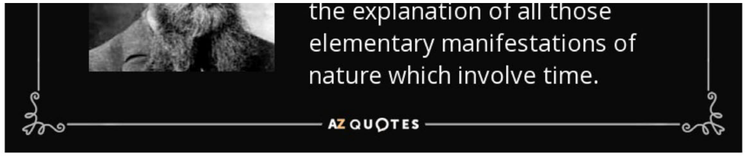
99.500 Applied Mathematics for Engineering

Spring Semester AY2026

Ordinary Differential Equation

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN





Week 3 Outline

- 1st-order linear ODE
- 2nd-order linear ODE
- Homogeneous vs non-homogeneous ODE
- Particular solution
- Mid-term project instruction

Definitions

$\frac{dy}{dx}$ $\frac{d^2y}{dx^2}$ Indep.

ODE: a relation containing one real variable x, the real dependent variable y, and some of its derivatives $y', y'', \cdots, y^{(n)}, \cdots$, with respect to x. The order of an ODE: the order of the highest derivative that occurs in the equation. Thus, an n-th order ODE has the general form

Order n.

$$F(x, y, y', y'', \ldots, \boxed{y^{(n)}}) = 0$$

$y \cdot y'$ X $y' \cdot y''$ X $(y'')^2$ ✓

Linear ODE: the dependent variable y and all its derivatives appear to the power of one (are not multiplied or divided by each other) and are not composed with any non-linear functions of y or its derivatives. The general form of an n n-th order linear ODE is:

(1) (2) $\sin y$ X $e^{y'}$ X

derivative

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

(x^2+1) $\frac{d^2y}{dx^2}$

If $g(x) \equiv 0$, it is a **homogeneous** equation.
If $g(x) \not\equiv 0$, it is a **non-homogeneous** equation.

Condition for ODE solution

The exact solution of n-th order ODE requires n "constraints":

1. Condition 1: If the independent variable is time, then the "constraints" are the initial conditions (ICs).
2. Condition 2: the independent variable is the position or location related, then the "constraints" are the boundary conditions (BCs).

1st-order linear ODE

to one side

rest other side

General form of a 1st-order linear ODE: $a_1(x)y' + a_0(x)y = g(x)$

$$a_1(x)y' + a_0(x)y = \boxed{0} \implies \frac{d}{dx}\; y' = -\frac{a_0(x)}{a_1(x)}y$$

$g(x)=0$ Homogeneous case

$$\implies \int \frac{dy}{y} = \int -\frac{a_0(x)}{a_1(x)}dx$$

$$\implies \ln|y| = \int -\frac{a_0(x)}{a_1(x)}dx$$

$$\implies y = Ce^{\int -\frac{a_0(x)}{a_1(x)}dx}$$

Non-homogeneous ODE: Integrating Factor

$$a_1(x)y' + \underbrace{\frac{a_0(x)}{a_1(x)}}_{P(x)}y = \underbrace{\frac{g(x)}{a_1(x)}}_{=Q(x)} \implies \boxed{y' + P(x)y = Q(x)}$$

$uy' + uPy = uQ(x)$

Integrating factor : define $\mu(x) = e^{\int P(x)dx}$

$\rightarrow u' = \frac{du}{dx} = e^{\int p(x)dx} \cdot \frac{d}{dx}\int p\,dx = u \cdot p(x)$

$= uy' + yu' = uy' + y\,u \cdot p(x)$

$$\Rightarrow \frac{d}{dx}[\mu(x)y] = \mu(x)y' + \mu(x)P(x)y \qquad \mu(x)y' + \mu(x)P(x)y = \mu(x)Q(x)$$

$$\implies \int \frac{d}{dx}[\mu(x)y] = \int \mu(x)Q(x)$$

$$\int \frac{d}{dx}[\mu(x)y]dx = \int \mu(x)Q(x)dx \qquad \implies \mu(x)y = \int \mu(x)Q(x)dx + C$$

$$\implies y = \frac{1}{\boxed{\mu(x)}}\left(\int \boxed{\mu(x)}Q(x)dx + C\right)$$

with $e^{\int P(x)dx}$

The solution to the 1st-order linear ODE $y' + P(x)y = Q(x)$ is:

$$y = \frac{1}{e^{\int P(x)dx}}\left(\int e^{\int P(x)dx}Q(x)dx + C\right)$$

Example

P Q $\int \frac{d}{dx}(e^{2x}y) = \int e^{3x}dx$

2x 2x

$$y' + 2y = e^x \qquad e^{\cdot}y = \tfrac{1}{3}e^{\cdot\cdot} + C$$

Identify $P(x)$ & $Q(x)$: $P(x) = 2$ & $Q(x) = e^x$

Find the integrating factor $\mu(x)$: $\mu(x) = e^{\int 2dx} = e^{2x}$

Multiply the ODE by $\mu(x)$: $\boxed{e^{2x}y' + 2e^{2x}y} = e^{2x}e^x$ divide by e^{2x}

Recognize the left-hand side as a derivative : $\frac{d}{dx}(e^{2x}y) = e^{3x}$

Integrate both sides : $e^{2x}y = \int e^{3x}dx + C = \frac{1}{3}e^{3x} + C$

Solve for y : $y = \frac{1}{3}e^x + Ce^{-2x}$

Apply Formula: $y = \frac{1}{e^{2x}}\left(\int e^{2x}\cdot e^{x}\,dx + C\right)$

$\frac{1}{e^{2x}}\left(\frac{1}{3}e^{3x} + C\right)$

2nd-order ODE: Method of Reduction to 1st Order

General form of a 2nd-order ODE : $\boxed{y''} = f(x, y, y')$

no y inside f $\longrightarrow$ $y'' = f(x, y')$ $\rightsquigarrow$ $P(x)y' + Q(x)$

$\boxed{y \text{ does not appear explicitly}}$: $y'' + P(x)y' = Q(x)$

let $\nu = y'$ & $v' = y''$ $\implies$ $\boxed{v' + P(x)v = Q(x)}$ 1st order ODE. by slide 7.

Method of integrating factor $\implies$ $v = \frac{1}{e^{\int P(x)dx}}\left(\int e^{\int P(x)dx}Q(x)dx + C\right)$

$\frac{d}{dx}v = \frac{d}{dx}y'$ $\implies$ $y' = \frac{1}{e^{\int P(x)dx}}\left(\int e^{\int P(x)dx}Q(x)dx + C\right)$

$\Downarrow$

$v' = y''$ $\implies$ $y = \int \frac{1}{e^{\int P(x)dx}}\left(\int e^{\int P(x)dx}Q(x)dx + C\right)dx$

Example

$\mu(x) = ?$

v' $\quad v$ $\qquad v = \frac{1}{u(x)}\left(\int u(x)Q(x)dx + c\right)$

$$y'' - 3y' = x$$

$v = y'$ & $v' = y''$ $\implies$ $\boxed{v' - 3v = x}$

Method of integrating factor $\implies$ $v = -\frac{1}{3}x - \frac{1}{9} + Ce^{3x}$ ✓

$\implies$ $\frac{dy}{dx}$ $y = \int -\frac{1}{3}x - \frac{1}{9} + Ce^{3x}dx$

$$y = -\frac{1}{6}x^2 - \frac{1}{9}x + Ce^{3x} + D$$

2nd-order ODE: Method of Reduction to 1st Order

$$y'' = f(\cancel{x}, y, y')$$

no x inside $\qquad = v \cdot \frac{dv}{dy}$

$\boxed{x \text{ does not appear explicitly}}$: $y'' = f(y, y')$

apply $\frac{d}{dx}$ $\qquad$ chain rule concept. $\qquad \frac{dy}{dx} = y' = v$

let $v = y'$ $\implies$ $y'' = \boxed{\frac{dv}{dx}} = \boxed{\frac{dv}{dy}\frac{dy}{dx}} = v\frac{dv}{dy}$ $\qquad$ with v & y.

$\implies$ $v\frac{dv}{dy} = f(y, v)$ 1st-order ODE

Find v $\implies$ $y = \int v dx$

Example

$\Rightarrow y'' = -y = f(y, y')$

$\lambda^2 + 1 = 0$

$\lambda = \pm i$

$y = C\cancel{e^{0x}}(\cos x) + D\cancel{e^{0x}}\sin(x)$

$$y'' + y = 0$$

let $v = y' \implies v\frac{dv}{dy} = -y$

$\implies \int v\,dv = -\int y\,dy$ $\quad \frac{v^2}{2} = -\frac{y^2}{2} + D$

$\implies v^2 = -y^2 + C$

$\implies \frac{dy}{dx} = \boxed{v} = \pm\sqrt{C - y^2}$

$\implies \frac{dy}{dx} = \pm\sqrt{C - y^2} \Rightarrow \int \frac{1}{\sqrt{C-y^2}}\,dy = \int \pm\,dx$

$\sin^{-1}\frac{y}{\sqrt{C}} = A - x,\ B + x$

$\implies y = \sqrt{C}\sin(A - x) + \sqrt{C}\sin(B + x)$

$\implies y = c_1 \sin(x) + c_2 \cos(x)$

+ve $\to y_1$, -ve $\to y_2$, combine $z = y_1 + y_2$

$y_1'' + y_1 = 0$ ✓

$y_2'' + y_2 = 0$ ✓

$z'' + z = (y_1 + y_2)'' + y_1 + y_2$

$= \underbrace{y_1'' + y_1}_{0} + \underbrace{y_2'' + y_2}_{0}$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A.$

$\frac{y}{\sqrt{C}} = \sin(A - x)$, $\sin(B + x)$

$y = \sqrt{C}\sin(A - x) + \sqrt{2}\sin(B + x)$

2nd-order Linear Homogeneous ODE with Constant Coefficients

$\lambda^2 \cancel{e^{\lambda x}} + a\lambda \cancel{e^{\lambda x}} + b\cancel{e^{\lambda x}} = 0$

$$y'' + ay' + by = 0$$

$(e^{\lambda x})'' = \lambda^2 e^{\lambda x}$

Look for solution inside of the form $y = e^{\lambda x}$ $\quad (e^{\lambda x})' = \lambda e^{\lambda x}$

$e^{\lambda x}$ is a solution $\iff \boxed{\lambda^2 + a\lambda + b = 0} \to$ characteristic eq

$\lambda_1 = \frac{1}{2}(-a + \sqrt{a^2 - 4b})$ solve

$\lambda_2 = \frac{1}{2}(-a - \sqrt{a^2 - 4b})$

Case 1 $a^2 - 4b > 0 \implies y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$

Case 2 $a^2 - 4b = 0 \implies \lambda_1 = \lambda_2, y = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$

$a^2 - 4b < 0 \implies \lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta$

$\implies y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$ $\quad e^{iz} = \cos z + i\sin z$

$\implies y = e^{\alpha x}(c_1 e^{i\beta x} + c_2 e^{-i\beta x})$

$\implies y = e^{\alpha x}(c_1 \cos(\beta x) + ic_1 \sin(\beta x) + c_2 \cos(\beta x) - ic_2 \sin(\beta x))$

$\implies y = Ce^{\alpha x}\cos(\beta x) + De^{\alpha x}\sin(\beta x)$

Examples

$y'' + y' - 2y = 0, y(0) = 4, y'(0) = -5$ Solution: $\lambda_1 = 1, \lambda_2 = -2$

$y = e^x + 3e^{-2x}$

$\lambda^2 + \lambda - 2 = 0$ BCs

$\lambda_1 = 1, \lambda_2 = -2$ $y(0) = C_1 + C_2 = 4$

$y = C_1 e^x + C_2 e^{-2x}$ $y'(0) = C_1 e^0 + C_2(-2)e^0 = -5$ $\Rightarrow C_1 = 1$, $C_2 = 3$

$C_1 - 2C_2 = -5$

$y'' - 4y' + 4y = 0, y(0) = 3, y'(0) = 1$ Solution: $\lambda_1 = \lambda_2 = 2$

$\lambda^2 - 4\lambda + 4 = 0$ $y = C_1 e^{2x} + C_2 x e^{2x}$ $\rightarrow y = (3 - 5x)e^{2x}$

$(\lambda - 2)^2 = 0$ $y(0) = C_1 + 0 = 3 \Rightarrow C_1 = 3$

$\lambda_1 = 2, \lambda_2 = 2$

$y'(0) = 2C_1 e^{2x} + C_2 e^{2x} + C_2 x 2 e^{2x}\big|_{x=0} = 2\cdot 3 + C_2 = 1$

$y'' - 2y' + 10y = 0$ Solution: $\lambda_1 = 1 + 3i, \lambda_2 = 1 - 3i$ $6 + C_2 = 1$

$\lambda^2 - 2\lambda + 10 = 0 \xRightarrow{\text{solve}} \lambda_1 = 1 + 3i$ $y = e^x(c_1 \cos 3x + c_2 \sin 3x)$ $C_2 = -5$

$\lambda_2 = 1 - 3i$ $y = Ce^{\alpha x}\cos(\beta x) + De^{\alpha x}\sin(\beta x)$

$\alpha = 1$ $\beta = 3$

Example of Reaction in Batch Reactor

System of liner ODE

$$\begin{cases} \frac{d\tilde{C}_R}{d\tilde{t}} &= -\tilde{C}_R + K_1\tilde{C}_I \\ \frac{d\tilde{C}_I}{d\tilde{t}} &= \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \\ \frac{d\tilde{C}_P}{d\tilde{t}} &= K_2\tilde{C}_I \end{cases} \text{with BCs} \begin{cases} \tilde{C}_R(0) &= 1 \\ \tilde{C}_I(0) &= 0 \\ \tilde{C}_P(0) &= 0 \end{cases}$$

$R \rightleftharpoons I \rightarrow P$

$\frac{d\tilde{C}_R}{d\tilde{t}} = \frac{d^2\tilde{C}_I}{d\tilde{t}} + K_1\frac{d\tilde{C}_I}{d\tilde{t}} + K_2\frac{d\tilde{C}_I}{d\tilde{t}}$

start with I

$$\frac{d\tilde{C}_I}{d\tilde{t}} = \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \implies \tilde{C}_R = \frac{d\tilde{C}_I}{d\tilde{t}} + K_1\tilde{C}_I + K_2\tilde{C}_I \quad (*)$$

2nd ODE Substitute $\tilde{C}_R$ into 1st ODE : $\frac{d^2\tilde{C}_I}{d\tilde{t}^2} + (1 + K_1 + K_2)\frac{d\tilde{C}_I}{d\tilde{t}} + K_2\tilde{C}_I = 0$

$$\lambda^2 + (1 + K_1 + K_2)\lambda + K_2 = 0$$

$$\lambda_{1,2} = \frac{-(1+K_1+K_2) \pm \sqrt{(1+K_1+K_2)^2 - 4K_2}}{2}$$

$$\tilde{C}_I = c_1 e^{\lambda_1 \tilde{t}} + c_2 e^{\lambda_2 \tilde{t}} \Rightarrow \frac{d\tilde{C}_I}{d\tilde{t}} = c_1\lambda_1 e^{0} + c_2\lambda_2 e^{0}$$

$$\tilde{C}_I(0) = c_1 + c_2 = 0$$

$$\boxed{\tilde{C}_R(0)} = \frac{d\tilde{C}_I}{d\tilde{t}}\bigg|_{\tilde{t}=0} = c_1\lambda_1 + c_2\lambda_2 = 1$$

$\frac{d\tilde{C}_I}{d\tilde{t}}\Big|_{\tilde{t}=0} + K_1\cancel{\tilde{C}_I(0)} + K_2\cancel{\tilde{C}_I(0)}$

$$c_1 = \frac{1}{\lambda_1 - \lambda_2} \ \& \ c_2 = \frac{1}{\lambda_2 - \lambda_1}$$

solve.

Continue

$$\tilde{C}_I = \frac{e^{\lambda_1\tilde{t}} - e^{\lambda_2\tilde{t}}}{\lambda_1-\lambda_2} \qquad \frac{d\tilde{C}_I}{d\tilde{t}}$$

$$\tilde{C}_R = \boxed{\frac{\lambda_1 e^{\lambda_1\tilde{t}} - \lambda_2 e^{\lambda_2\tilde{t}}}{\lambda_1-\lambda_2}} + (K_1+K_2)\boxed{\frac{e^{\lambda_1\tilde{t}} - e^{\lambda_2\tilde{t}}}{\lambda_1-\lambda_2}}$$

$\tilde{C}_I$

$$\begin{aligned}
\int \frac{d\tilde{C}_P}{d\tilde{t}} &= K_2\tilde{C}_I = \frac{K_2\left(e^{\lambda_1\tilde{t}} - e^{\lambda_2\tilde{t}}\right)}{\lambda_1-\lambda_2} \\
\tilde{C}_P &= \frac{K_2}{\lambda_1-\lambda_2}\int\left(e^{\lambda_1\tilde{t}} - e^{\lambda_2\tilde{t}}\right)d\tilde{t} \\
&= \frac{K_2}{\lambda_1-\lambda_2}\left(\frac{e^{\lambda_1\tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2\tilde{t}}}{\lambda_2}\right) + c_3
\end{aligned}$$

$\tilde{t}=0,\ \tilde{C}_P(0)=0$

$$\begin{aligned}
\tilde{C}_P(0) &= \frac{K_2}{\lambda_1-\lambda_2}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) + c_3 = 0 \\
&\Longrightarrow c_3 = \frac{K_2}{\lambda_1\lambda_2} \\
\tilde{C}_P &= \frac{K_2}{\lambda_1-\lambda_2}\left(\frac{e^{\lambda_1\tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2\tilde{t}}}{\lambda_2}\right) + \frac{K_2}{\lambda_1\lambda_2}
\end{aligned}$$

Nth-order Linear Homogeneous ODE with Constant Coefficients

General form : $y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1}y' + a_n y = \boxed{0}$ ← Homo case.

if $y = e^{\lambda x}$ is a solution $\iff \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0$ characteristic eqns

λ : characteristic valus or eigenvalues ↙ solve

$$\lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = (\lambda-\lambda_1)^{m_1}(\lambda-\lambda_2)^{m_2}\cdots(\lambda-\lambda_s)^{m_s}$$

$\lambda_1, \cdots, \lambda_s$: distinct eigenvalues

$m_1, \cdots, m_s$: multiplicity of respective eigenvaluess

Theorem: Let $\lambda_1, \cdots, \lambda_s$ be the distinct eigenvalues for nth-order linear homogeneous ODE, with multiplicity $m_1, \cdots, m_s$ respectively. Then it has a fundamental set of solutions

if $m_1 = 3$. $e^{2x}, xe^{2x}, x^2e^{2x}$

for $\lambda_1 \to e^{\lambda_1 x}, xe^{\lambda_1 x}, \cdots, x^{m_1-1}e^{\lambda_1 x}$; $x^2e^{\lambda_1 x}$ ..

for $\lambda_2 \to e^{\lambda_2 x}, xe^{\lambda_2 x}, \cdots, x^{m_2-1}e^{\lambda_2 x}$ ....;

$(\lambda-2)^3(\lambda-3)^4$

$\lambda_1 = 2,\ m_1 = 3$

for $\lambda_s \to e^{\lambda_s x}, xe^{\lambda_s x}, \cdots, x^{m_s-1}e^{\lambda_s x}$.

$\lambda_2 = 3,\ m_2 = 4$

The general soluton for the homogeneous ODE is the linear combination of fundamental set of solutions: $y_h(x) = \sum_{i=1}^{s}\sum_{j=1}^{m_i} c_{ij}x^{j-1}e^{\lambda_i x}$

Nth-order Linear Non-homogeneous ODE

non-Homo

$$\boxed{y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y} = \boxed{f(x)}$$

The general solution is in the form: $y(x) = y_h(x) + y_p(x)$

homogeneous solution $y_h(x)$: $y_h^{(n)} + a_1(x)y_h^{(n-1)} + \cdots + a_{n-1}(x)y_h' + a_n(x)y_h = 0$

particular solution $y_p(x)$: $y_p^{(n)} + a_1(x)y_p^{(n-1)} + \cdots + a_{n-1}(x)y_p' + a_n(x)y_p = f(x)$

How to find the particular solution?

1. Method of variation of parameters
2. Method of undetermined coefficients

Method of Variation of Parameters

$u_1'y_1' + u_1y_1'' + u_2'y_2' + u_2y_2''$
$+ a(u_1y_1' + u_2y_2')$
$+ b(u_1y_1 + u_2y_2) = f$

(*) $y_p'' = \frac{d}{dx}(u_1y_1' + u_2y_2') = u_1'y_1' + u_1y_1'' + u_2'y_2' + u_2y_2''$

$$y'' + ay' + by = f(x)$$

set 0

Let y_1 & y_2 be two linearly independent homogeneous solutions

Look for $y_p = u_1y_1 + u_2y_2$ ($u_1(x)$, $u_2(x)$) $\Rightarrow y_p' = u_1'y_1 + u_1y_1' + u_2'y_2 + u_2y_2'$

Suppose $u_1'y_1 + u_2'y_2 = 0$ $\quad = y_p' = u_1y_1' + u_2y_2'$ (*)

$y_p'' + a y_p' + b y_p = f$

Put y_p into $y'' + ay' + by = f \Longrightarrow \boxed{u_1'y_1' + u_2'y_2' = f}$

$$\begin{cases} u_1'y_1 + u_2'y_2 = 0 \\ u_1'y_1' + u_2'y_2' = f \end{cases} \Longrightarrow u_1' = -\frac{y_2 f}{y_1y_2' - y_1'y_2},\ u_2' = \frac{y_1 f}{y_1y_2' - y_1'y_2}$$

$$u_1 = -\int \frac{y_2 f}{y_1y_2' - y_1'y_2}dx$$

$$u_2 = \int \frac{y_1 f}{y_1y_2' - y_1'y_2}dx$$

Example

y_1 is soln to $y''+y=0$
y_2 is " to $y''+y=0$

$$y'' + y = \sec x \quad f$$

$$\boxed{y_1 = \cos x, y_2 = \sin x} \quad \text{Homo Soltn.}$$

$$y_1 y_2' - y_1' y_2 = \cos x \cos x - (-\sin x)\sin x = 1$$

$$u_1 = -\int \underbrace{\sin x}_{y_2} \underbrace{\sec x}_{f}\, dx = \ln|\cos x| + c_1$$

$$u_2 = \int \underbrace{\cos x}_{y_1} \underbrace{\sec x}_{f}\, dx = x + c_2$$

$$y_p = u_1 y_1 + u_2 y_2 \qquad y_p = \underbrace{\cos x}_{y_1} \underbrace{\ln|\cos x|}_{u_1} + \underbrace{x}_{u_2} \underbrace{\sin x}_{y_2}$$

$$y = \underbrace{c_1 \cos x + c_2 \sin x}_{\text{Homo}} + \underbrace{\cos x \ln|\cos x| + x \sin x}_{\text{parti}}$$

$$u_1' y_1 + u_2' y_2 = \frac{-\sin x}{\cos x} \cdot \cos x + 1 \cdot \sin x = 0.$$

Method of Undetermined Coefficients: Case 1

① find Homo part.

$$\boxed{y'' + by' + cy} = f(x) \qquad f(x) = P_n(x)e^{\alpha x}$$

$$f(x) = P_n(x)e^{\alpha x}$$

$P_n(x)$: polynomial of degree $n \geq 0$

$$y_p' = Q'(x)\cancel{e^{\alpha x}} + Q(x)\alpha \cancel{e^{\alpha x}}$$

$$y_p'' = Q''(x)\cancel{e^{\alpha x}} + Q'(x)\alpha\cancel{e^{\alpha x}} + \alpha Q'(x)\cancel{e^{\alpha x}} + \alpha^2 Q(x)\cancel{e^{\alpha x}}$$

Look for $y_p = Q(x)e^{\alpha x}$

$$Q'' + (2\alpha + b)Q' + (\alpha^2 + b\alpha + c)Q = P_n(x)$$

poly of degree n to be determined.

S1 $\alpha^2 + b\alpha + c \neq 0$ Choose $Q(x) = R_n(x), y = R_n(x)e^{\alpha x}$

S2 $\alpha^2 + b\alpha + c = 0, 2\alpha + b \neq 0$ Choose $Q(x) = xR_n(x), y = xR_n(x)e^{\alpha x}$

S3 $\alpha^2 + b\alpha + c = 0, 2\alpha + b = 0$ Choose $Q(x) = x^2R_n(x), y = x^2R_n(x)e^{\alpha x}$

Examples

$y_h = c_1 e^{2x} + c_2 e^{-x}$

$\lambda^2 - \lambda - 2 = 0,\ (\lambda-2)(\lambda+1)=0 \Rightarrow \lambda = 2, \lambda = -1.$

$$y'' - y' - 2y = 4x^2$$ $a=0,\ b=-1,\ c=-2$

Solution: $y = c_1 e^{2x} + c_2 e^{-x} - 3 + 2x - 2x^2$ $\alpha^2 + b\alpha + c = 0+0+(-2) \neq 0$ (S1)

Choose $Q(x) = R_2(x) \Rightarrow y_p = R_2(x)\cdot e^{0x} = Ax^2 + Bx + C \Rightarrow y_p' = 2Ax + B$

$y_p'' = 2A.$

put y_p, y_p', y_p'' into original eq:

$$\underbrace{2A}_{y_p''} - (\underbrace{2Ax+B}_{y_p'}) \underbrace{-2(Ax^2+Bx+C)}_{-2y_p} = 4x^2$$

Compare coeff: (x^0) $2A - B - 2C = 0 \Rightarrow C = -3$

(x^1) $-2Ax - 2Bx = 0 \Rightarrow B = 2$

(x^2) $-2Ax^2 = 4x^2 \Rightarrow A = -2$

$\Rightarrow y_p = -2x^2 + 2x - 3$

Examples

$$y''' + 2y'' - y' = 3x^2 - 2x + 1$$ $a=0,\ b=2,\ c=-1,\ \alpha^2 + a\beta + c \neq 0$ (S1)

Solution: $y = c_1 + c_2 e^{(-1+\sqrt{2})x} + c_3 e^{(-1-\sqrt{2})x} - 27x - 5x^2 - x^3$

Let $z = y' \Rightarrow z'' + 2z' - z = 3x^2 - 2x + 1$

$\lambda^2 + 2\lambda - 1 = 0 \Rightarrow \lambda = \frac{-2 \pm \sqrt{2^2+4}}{2} = -1 \pm \sqrt{2} \Rightarrow z_h = c_1 e^{(-1-\sqrt{2})x} + c_2 e^{(-1+\sqrt{2})x}$

$z_p = Ax^2 + Bx + C,\ z_p' = 2Ax + B,\ z_p'' = 2A.$

Put z_p, z_p', z_p'' into original eq: $2A + 2(2Ax+B) - (Ax^2 + Bx + C) = 3x^2 - 2x + 1$

Compare coeff

$(x^2): -Ax^2 = 3x^2 \Rightarrow A = -3$

$(x^1): 4Ax - Bx = -2x \Rightarrow 4A - B = -2 \Rightarrow B = -12+2 = \ldots$

$(x^0): 2A + 2B - C = 1 \Rightarrow C = 2A + 2B - 1$

$= -6 - 20 - 1 = -27.$

$z_p = -3x^2 - 10x - 27.$

$z = z_h + z_p = c_1 e^{(-1-\sqrt{2})\ldots}$

$y = \int z\,dx = c_1 e^{(-1-\sqrt{\ldots}}$

Examples

$P_n(x)\ n=1.$

$$y'' - 2y' + y = xe^x$$

Solution: $y = c_1 e^x + c_2 x e^x + \frac{1}{6}x^3 e^x$

$\lambda^2 - 2\lambda + 1 = 0,\ (\lambda-1)^2 = 0\ \lambda = 1$ or $2.$

$y_h = c_1 e^x + c_2 x e^x$

$\alpha = 1,\ b = -2,\ c = 1.\quad \alpha^2 + b\alpha + c = 0$ (S3)

$Q(x) = x^2(Ax+B)\quad y_p = x^2(Ax+B)e^x$

$y_p' = (3Ax^2 + 2Bx)e^x + y_p,\ y_p'' = (3Ax^2 + 2Bx)e^x + (6Ax + 2B)e^x + y_p'$

put y_p, y_p', y_p'' into original eqn let $= xe^x$. $\boxed{\text{compare coefficient.}}$ to obtain $A = \frac{1}{6}, B = 0$

$y_p = x^2 \cdot \frac{1}{6} x e^x = \frac{1}{6}x^3 e^x$

Method of Undetermined Coefficients: Case 2

$e^{\alpha+i\beta} = e^{\alpha}\boxed{e^{i\beta}} = e^{\alpha}\boxed{(\cos\beta x + i\sin\beta x)}$

$y'' + by' + cy = f(x)\quad f(x) = P_n(x)e^{\alpha x}\cos(\beta x)$ or $f(x) = P_n(x)e^{\alpha x}\sin(\beta x)$

Look for y_p s.t. $\quad y'' + by' + cy = \boxed{P_n(x)e^{(\alpha+i\beta)x}}\ [*]$

Using method in Case 1 to obtain: $\quad z(x) = \underbrace{u(x)}_{\text{Real}} + iv(x) \rightarrow$ imaginary

Substitute $z(x)$ into $[*]$ $\quad u(x)$ is solution of $y'' + by' + cy = P_n(x)e^{\alpha x}\cos(\beta x)$

$u(x) + iv(x)$ $\quad v(x)$ is solution of $y'' + by' + cy = P_n(x)e^{\alpha x}\sin(\beta x)$

$\alpha = 1\quad \beta = 1$

look for y_p'' s.t. $y_p'' \ldots$

$y_p'' - 2y_p' + 2y_p = P_n(x)\, e^{(1+i)x} = e^0 \cdot e^{\ldots} = e^i e^x = e^{(1+i)x}$

$\rightarrow$ poly of order 0, which is const.

check $\gamma^2 + b\gamma + c = (1+i)^2 + (-2)(1+i) + 2$

$\gamma = 1+i$

$b = -2$ $\quad = 1 - 1 + 2i - 2 - 2i + 2 = 0$

$c = 2$ $\quad 2\gamma + b = 2(1+i) - 2 \neq 0$ $\boxed{S2} \Rightarrow y_p = x P_n(x) e^{\gamma x} = A x e^{\gamma x}$

Method of Undetermined Coefficients: Case 2

$y_h = c_1 e^x \cos x + c_2 e^x \sin x$

$\lambda^2 - 2\lambda + 2 = 0$, $\lambda^2 - 2\lambda + 1 = -1$, $(\lambda - 1)^2 = -1$, $\lambda - 1 = \pm i$

Example $\quad y'' - 2y' + 2y = e^x \cos x = 1 \cdot e^x \cos x$

$$y = c_1 e^x \cos(x) + c_2 e^x \sin(x) + \frac{x e^x \sin x}{2}$$

$y_p' = Ax\gamma e^{\gamma x} + A e^{\gamma x} = \gamma y_p + A e^{\gamma x}$

$y_p'' = \gamma y_p' + A\gamma e^{\gamma x}$

$y_p'' - 2y_p' + 2y_p = e^{\gamma x}$

$\gamma y_p' + A\gamma e^{\gamma x} - 2y_p' + 2y_p$

$= (\gamma - 2) y_p' + A\gamma e^{\gamma x} + 2y_p$

$= (\gamma - 2)(\gamma y_p + A e^{\gamma x}) + A\gamma e^{\gamma x} + 2y_p$

$= \gamma^2 y_p + A\gamma e^{\gamma x} - 2\gamma y_p - 2A e^{\gamma x} + A\gamma e^{\gamma x} + 2y_p$

$\gamma^2 y_p - 2\gamma y_p + 2y_p = [(1+i)^2 - 2(1+i) + 2]\, y_p = 0$

$1 + (-1) + 2i - 2 - 2i + 2 = 0$

Sum

$(2A\gamma - 2A) e^{\gamma x} = e^{\gamma x}$

$2A(1+i) - 2A = 1$

$2Ai = 1 \Rightarrow A = \frac{1}{2i} = \frac{i}{2i^2} = -\frac{1}{2} i$

Method of Undetermined Coefficients: Case 2

Example $\quad y'' - 2y' + 2y = e^x \cos x$

$$y = c_1 e^x \cos(x) + c_2 e^x \sin(x) + \frac{x e^x \sin x}{2}$$

$y_p = -\frac{1}{2} i x e^{(1+i)x} = -\frac{1}{2} i x e^x (\cos x + i \sin x)$ $\rightarrow$ take $i \sin x$ part.

take real part $= -\frac{1}{2} i x e^x i \sin x$

$$= \frac{1}{2} x e^{x} \sin x$$

$$y = y_h + y_p = c_1 e^{x} \cos x + c_2 e^{x} \sin x + \frac{1}{2} x e^{x} \sin x$$

Announcements

- No homework this week
- Homework for week 2, 5, 9, 10, 11 with each 2% for the final grade
- Read “Guidelines for Mid-term Project” in eDimension
- Next week CNY: No class